Solution Guide for D84361GC10 Build Composite Applications

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Quick Reference: Setting the Start or End State for a Practice

Each practice follows the general steps on this summary page. To set the start or end state for a given practice, it is recommended that you follow the specific instructions for that practice later in this document.

1. If the desired practice has already been deployed, undeploy the application(s).
2. If the desired practice has already been created, rename or delete the directory containing that project.
3. To set to the end state for a given practice:
    a. Complete steps 1-2 above.
    b. Be sure practice 1 has been completed (manually).
    c. Complete any other manual steps required for the project.
    d. Copy the solution for the desired practice to the appropriate directory.
    e. Deploy and test the project.

This practice	…deploys these projects:	To set the start state, complete Practice 1 and…	
		Install the solution file for this practice(s)	Open and deploy these applications.
1. Overview	none		
2. HelloWorld	none		
3. Managing Applications	Basics\HelloWorld		
4. Working With Data	Basics\NativeData		
5. Adapters	Basics\CCValidate Basics\JMSProducer Basics\JMSConsumer [optional]		
6. Hello BPEL	BPELProjects\HelloBPEL	practice 05	CCValidate
7. More BPEL	BPELProjects\AsyncDemo		
8. Fault Handling	BPELProjects\Enroll BPELProjects\Booking BPELProjects\CallGreeting BPELProjects\SayHello		
9. Human Task	BPELProjects\Approval BPELProjects\ApprovalTaskForm		
10. Reuse	BPELProjects\OrderFlow	practice 05 practice 09	CCValidate Approval ApprovalTaskForm
11. REST	BPELProjects\HelloBPEL (redeploy) BPELProjects\CallREST	practice 05	CCValidate
12. Business Rules	BPELProjects\OrderProcessing	practice 09	Approval ApprovalTaskForm
13. Event	BPELProjects\UpdateStock		

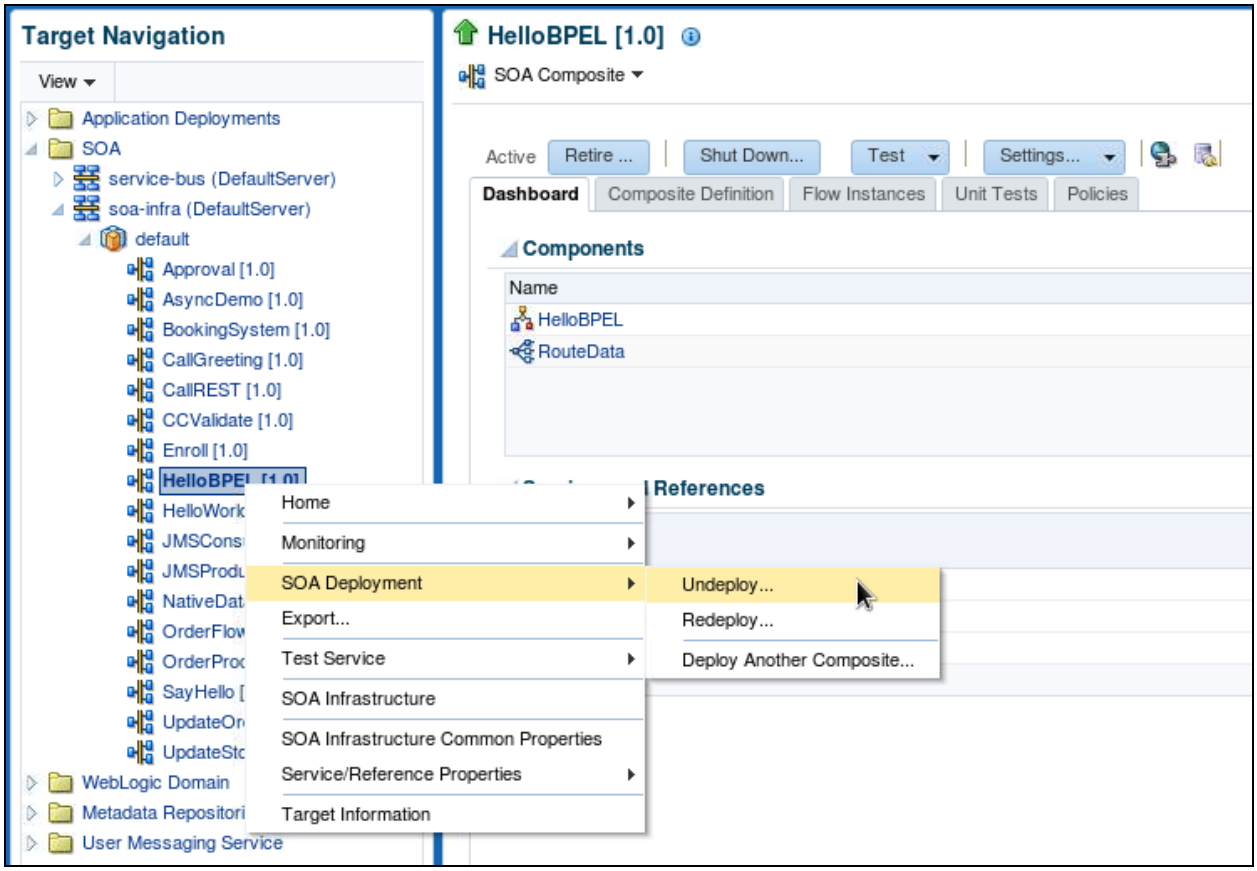
This practice	…deploys these projects:	To set the start state, complete Practice 1 and…	
		Install the solution file for this practice(s)	Open and deploy these applications.
	BPELProjects\UpdateOrder		
14. Testing	Basics\CCValidate (redeploy) BPELProjects\HelloBPEL (redeploy)	practice 05 practice 06	CCValidate HelloBPEL
15. Security	BPELProjects\Enroll (redeploy) BPELProjects\Booking (redeploy)	practice 08	Enroll Booking

Undeploying a Composite Application

Overview

If you wish to deploy a solution file for an application that has already been deployed, remove the deployed version from the runtime environment first.

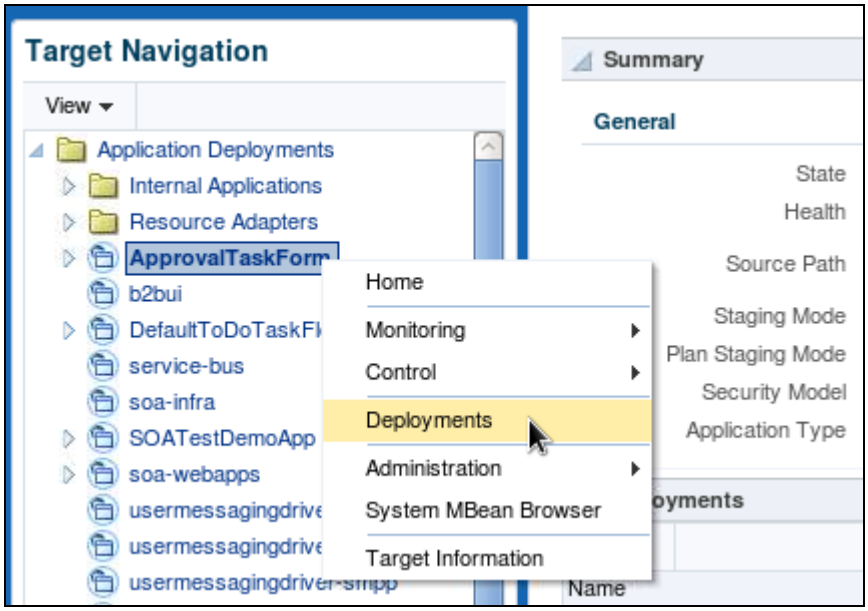
1. Undeploy the Composite Application.
   a. Open the Oracle Enterprise Manager page (`http://localhost:7101/em`).
   b. In the Target Navigation pane, expand the SOA > soa-infra > default nodes in the tree.
   c. Right-click the name of the application you wish to undeploy.
   d. Select SOA Deployment > Undeploy.



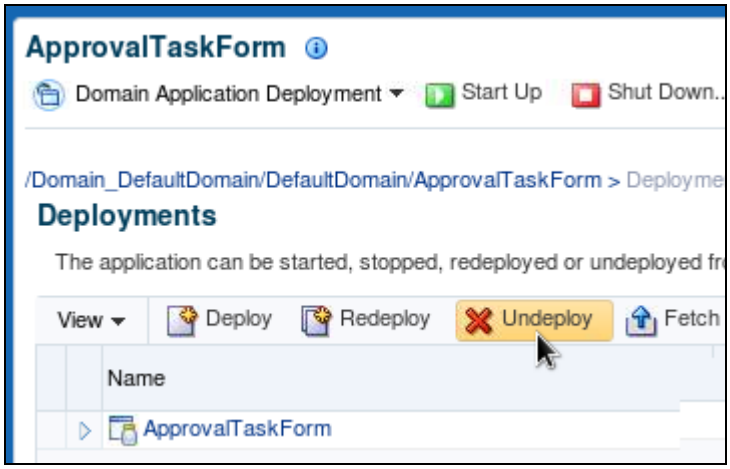
   The Confirmation screen is displayed.
   e. Click Undeploy.
   f. The Progress screen is displayed. When undeployment is complete, you are returned to the soa-infra page of the Enterprise Manager.
   g. Verify that the application is no longer displayed under the default node of the soa-infra branch of the Target Navigation pane.

Undeploying an Application (Practice 09 uses this.)

1. For Practice 9 ONLY, undeploy the application ApprovalTaskForm.
   a. In the Target Navigation pane, expand the Application Deployments branch of the tree.
   b. Right-click application `ApprovalTaskForm` and select Deployments.



   c. On the ApprovalTaskForm screen, click Undeploy.



   d. The Progress screen is displayed. When undeployment is complete, you are returned to the soa-infra page of the Enterprise Manager.
   e. Verify that the application is no longer displayed under the Application Deployments branch of the Target Navigation pane.

Installing a Solution File

Overview

To deploy a solution file, complete the instructions below.

Assumptions

These instructions assume that:

- You have removed any deployed version of the solution application from the runtime environment.
- All work is located under /u01/app/fmw12c/domains/mywork/<*ApplicationName*>. (*ApplicationName* will be either Basics or BPELProjects.)
- The solution files are located in /home/oracle/labs/soln/mywork/<*ApplicationName*>/<*projectName*>.

Tasks

1. If work on a given lab is partially complete, delete or rename the directory for that lab under /u01/app/fmw12c/domains/mywork. For instance, if project 5 has been partially completed, rename directory /u01/app/fmw12c/domains/mywork/Basics/CCValidate (for example, CCValidate.old).
2. Copy the solution directory /home/oracle/labs/soln/mywork/<*ApplicationName*>/<*projectName*> to /u01/app/fmw12c/domains/mywork/<*ApplicationName*>.
3. Open and deploy the project(s).

   Important Note: Refer to the specific deployment instructions in the Activity Guide for the solution you are deploying. Several of the lessons deploy multiple applications. Recall also that Practice 9 follows a different set of deployment steps than other practices.

Practice 01 Installing Oracle SOA Suite 12*c*

- This practice must be completed manually.
- It is a prerequisite for all other labs.

Practice 02 Hello World

Set the start state:

1. If it is not already done, complete Practice 01.
2. If project `HelloWorld` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/Basics/
     mv HelloWorld HelloWorld.old
     ```
   - If you do not wish to save your work:
     - Delete project `HelloWorld` from JDeveloper.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/Basics/HelloWorld
   /u01/app/fmw12c/domains/mywork/Basics.
   ```

Practice 03 Managing and Monitoring

Set the start state:

1. Complete the Set the end state instructions for Practice 02.

Set the end state:

- This practice must be completed manually.

Practice 04 Working With Data

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `Basics`.
3. If project `NativeData` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/Basics/
     mv NativeData NativeData.old
     ```
   - If you do not wish to save your work:
     - Delete project `NativeData` from JDeveloper.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/Basics/NativeData
   /u01/app/fmw12c/domains/mywork/Basics.
   ```
3. Deploy Project `NativeData` as described in Practice 4-5.

Practice 05 JDBC and JMS Adapters

This practice includes three separate projects. Some later projects depend on project `CCValidate`. There are no dependencies on project `JMSProducer`. Project `JMSConsumer` depends on `JMSProducer`.

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `Basics`.
3. If project `CCValidate` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/Basics/
     mv CCValidate CCValidate.old
     ```
   - If you do not wish to save your work:
     - Delete project `CCValidate` from JDeveloper.
4. Repeat step 3 for Projects `JMSProducer` and `JMSConsumer`.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Complete Practice 5-1 manually.
3. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/Basics/CCValidate
   /u01/app/fmw12c/domains/mywork/Basics.
   ```
4. Deploy and test project `CCValidate` as described in Practice 5-4.
5. Complete Practice 5-5 manually.
6. Deploy project `JMSProducer` as described in Practice 5-7.
7. Deploy project `JMSConsumer` as described in Practice 5-9 (NOT 5-8).

   Note: *The `JMSConsumer` solution project includes both Practices 5-8 and 5-9. If you choose to deploy and test `JMSConsumer`, you must provide the customer name as described in Practice 5-9, step 5.*

Practice 06 HelloBPEL

This practice assumes practice 05 has been built and deployed successfully.

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `BPELProjects`.
3. If it is not already built and successfully deployed, complete the *Set the end state* instructions for Practice 05 (project `CCValidate`).
4. If project `HelloBPEL` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/BPELProjects/
     mv HelloBPEL HelloBPEL.old
     ```
   - If you do not wish to save your work:
     - Delete project `HelloBPEL` from JDeveloper.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/BPELProjects/HelloBPEL
   /u01/app/fmw12c/domains/mywork/BPELProjects.
   ```
3. Deploy and test the project as described in Practice 6-2

Practice 07 More BPEL

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `BPELProjects`.
3. If project `AsyncDemo` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/BPELProjects/
     mv AsyncDemo AsyncDemo.old
     ```
   - If you do not wish to save your work:
     - Delete project `AsyncDemo` from JDeveloper.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/BPELProjects/AsyncDemo
   /u01/app/fmw12c/domains/mywork/BPELProjects
   ```
3. Deploy and test the project as described in Practice 7-4.

Practice 08-1 through 08-4 Fault Handling

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `BPELProjects`.
3. If project `Enroll` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/BPELProjects/
     mv Enroll Enroll.old
     ```
   - If you do not wish to save your work:
     - Delete project `Enroll` from JDeveloper.
4. Repeat step 3 for project `BookingSystem`.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/BPELProjects/Enroll
   /u01/app/fmw12c/domains/mywork/BPELProjects
   ```
3. Repeat step 2 for project `Booking`.
4. Deploy project `BookingSystem` as described in Practice 8-1.
5. Deploy and test project `Enroll` as described in Practice 8-4.

Practice 08-5 Fault Management Framework

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `BPELProjects`.
3. If project `SayHello` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/BPELProjects/
     mv SayHello SayHello.old
     ```
   - If you do not wish to save your work:
     - Delete project `SayHello` from JDeveloper.
4. Repeat step 3 for project `CallGreeting`.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/BPELProjects/SayHello
   /u01/app/fmw12c/domains/mywork/BPELProjects
   ```
3. Repeat step 2 for project `CallGreeting`.
4. Deploy and test project `SayHello` as described in Practice 8-5 task 1.
5. Deploy and test project `CallGreeting` as described in Practice 8-5 task 1.

Practice 09 Human Tasks

Because this Practice deploys an ADF application with the project, the restore instructions and solution files are different than other practices.

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `BPELProjects`.
3. If project `Approval` already exists:
   - If the project is already deployed:
     A. Undeploy project `Approval`.
     B. Undeploy application `ApprovalTaskForm`.

     Note: See instructions at the beginning of this document.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/BPELProjects/
     mv Approval Approval.old
     ```
   - If you do not wish to save your work:
     - Delete project `Approval` from JDeveloper.
4. Repeat step 3 for project `ApprovalTaskForm`.
5. Complete Practice 9-1 and 9-2 manually.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/BPELProjects/Approval
   /u01/app/fmw12c/domains/mywork/BPELProjects
   ```
3. Repeat step 2 for project `ApprovalTaskForm`.
4. Deploy project `Approval` as described in Practice 9-6. (These instructions include the deployment of `ApprovalTaskForm`.)

Practice 10 Sharing Functionality

This practice assumes that

- You have completed Practice 5 using *exactly* the project and component names specified in that practice, and that the `ValidateCreditCard` application is still deployed to the application server.
- You have completed Practice 9 using *exactly* the project name and component names specified in that practice, and that the `Approval` application is still deployed to the application server.

If your environment does not meet these assumptions, deploy and test the working solutions (set the end state) for the missing practice(s) as described in this document.

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `BPELProjects`.
3. If project `OrderFlow` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/BPELProjects/
     mv OrderFlow OrderFlow.old
     ```
   - If you do not wish to save your work:
     - Delete project `OrderFlow` from JDeveloper.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Complete Practice 10-1 step 1 manually.
3. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/BPELProjects/OrderFlow
   /u01/app/fmw12c/domains/mywork/BPELProjects
   ```
4. Deploy and test project `OrderFlow` as described in Practice 10-2.

Practice 11 REST Adapter

This practice assumes that you successfully created and deployed the project `CCValidate` in Practice 05 and project `HelloBPEL` in Practice 06.

If your environment does not meet these assumptions, deploy and test the working solutions (set the end state) for the missing practice(s) as described in this document.

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `BPELProjects`.
3. If project `HelloBPEL` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/BPELProjects/
     mv HelloBPEL HelloBPEL.old
     ```
   - If you do not wish to save your work:
     - Delete project `HelloBPEL` from JDeveloper.
4. Repeat step 3 for project `CallREST`.
5. If necessary, complete the Set the End State instructions for Practice 05.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/BPELProjects/HelloBPEL
   /u01/app/fmw12c/domains/mywork/BPELProjects
   ```
3. Repeat step 2 for project `CallREST`.
4. Deploy and test project `HelloBPEL` as described in Practice 6-2.
5. Deploy and test project CallREST as described in Practice 11-3.

Practice 12 Business Rules

This practice assumes that you have completed Practice 9 successfully, and have deployed the `Approvals` and `ApprovalTaskForm` applications.

Set the start state:

1. If it is not already done, complete Practice 01.
2. If necessary, complete the *Set the end state* instructions for Practice 9.
3. If project `OrderProcessing` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/BPELProjects/
     mv OrderProcessing OrderProcessing.old
     ```

   - If you do not wish to save your work:
     - Delete project `OrderProcessing` from JDeveloper.
4. If necessary, complete the *Set the End State* instructions for Practice 09.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/BPELProjects/OrderProcessing
   /u01/app/fmw12c/domains/mywork/BPELProjects
   ```

3. Deploy and test project `OrderProcessing` as described in Practice 12-3 and [optionally] test as described in Practice 12-6.

Practice 13 Events

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it does not exist, use JDeveloper to create application `BPELProjects`.
3. If project `UpdateStock` already exists:
   - If the project is already deployed, undeploy it.
   - If you wish to save your work, issue the following commands:

     ```
     cd /u01/app/fmw12c/domains/mywork/BPELProjects/
     mv UpdateStock UpdateStock.old
     ```
   - If you do not wish to save your work:
     - Delete project `UpdateStock` from JDeveloper.
4. Repeat step 2 for project `UpdateOrder`.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Complete Practice 13-1 step 1 manually.
3. Issue the following command to install the solution:

   ```
   cp -r /home/oracle/labs/soln/mywork/BPELProjects/UpdateStock
   /u01/app/fmw12c/domains/mywork/BPELProjects
   ```
4. Repeat step 2 for project `UpdateOrder`.
5. Deploy project `UpdateStock` as described in Practice 13-1 task 8.
6. Deploy project `UpdateOrder` as described in Practice 13-2 task 5.
7. To test the projects, complete Practice 13-3 manually.

Practice 14 Testing and Debugging

This practice assumes that you have successfully completed and deployed the `CCValidate` (Practice 05) and `HelloBPEL` (Practice 06) applications, which were for the JDBC Adapter and Introduction to BPEL lessons, respectively.

Set the start state:

1. Complete the *Set the end state* instructions for Practice 06.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Test project `CCValidate` as described in Practice 14-3 steps 2-6.
3. Debug project `HelloBPEL` as described in Practice 14-4 steps 4-15.

Practice 15 Security

This practice assumes that you have completed Practice 8 successfully and that projects `Enroll` and `BookingSystem` are deployed.

Set the start state:

1. If it is not already done, complete Practice 01.
2. If it is not already done, complete the *Set the end state* instructions for Practice 8.

Set the end state:

1. Complete the *Set the start state* instructions for this practice.
2. Complete this practice manually.